

Course: Logistic Regression

Binary classification, sigmoid and model tuning

K. Kadri

Outline

- 1 Intuition and role of logistic regression
- 2 Mathematical formulation
- 3 Regularization and hyperparameters
- 4 Thresholds and ROC visual
- 5 Advantages and limitations

Goal: predict a binary class

- Predict a variable $y \in \{0, 1\}$ (e.g., sick / healthy).
- Linear regression gives unbounded values.
- We need a model that outputs a **probability** in $[0, 1]$:

$$p(y = 1 \mid x)$$

Key idea

Logistic regression predicts a **probability**, then applies a classification **threshold**.

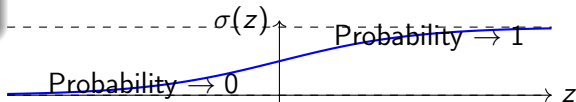
The sigmoid function (visual)

Definition

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid:

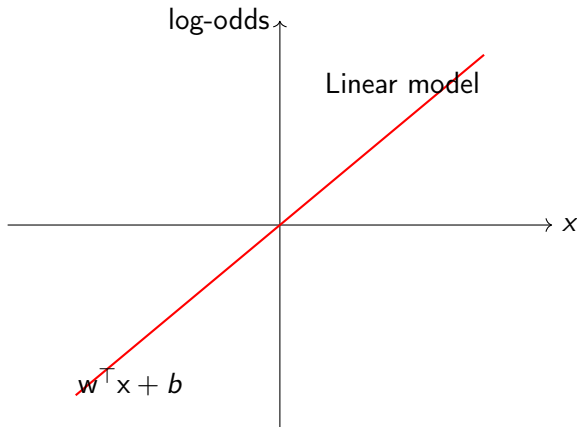
- maps $z \in \mathbb{R}$ to $[0, 1]$,
- is interpretable as a probability,
- produces an S-shaped curve.



Log-odds: linear view (visual)

$$\log\left(\frac{p}{1-p}\right) = w^\top x + b$$

- Logistic regression is linear in the **log-odds** space.
- Each coefficient w_j measures the effect of x_j .



Cost function: log-loss

For (x_i, y_i)

$$\mathcal{L}(y_i, \hat{p}_i) = -[y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)]$$

- Confident + correct $\rightarrow$ low loss.
- Confident + wrong $\rightarrow$ huge loss.

Goal

Minimize average log-loss.

Regularization (L1 / L2)

- Avoid overfitting by shrinking coefficients.

- L2:

$$\lambda \|w\|_2^2$$

- L1:

$$\lambda \|w\|_1$$

Effect

- L2 → smooth shrinking.
- L1 → promotes sparsity (feature selection).

Hyperparameter C in scikit-learn

$$C = \frac{1}{\lambda}$$

- Small $C \rightarrow$ strong regularization.
- Large $C \rightarrow$ weak regularization.

Tuning

Use cross-validation (GridSearch / RandomizedSearch).

Decision threshold

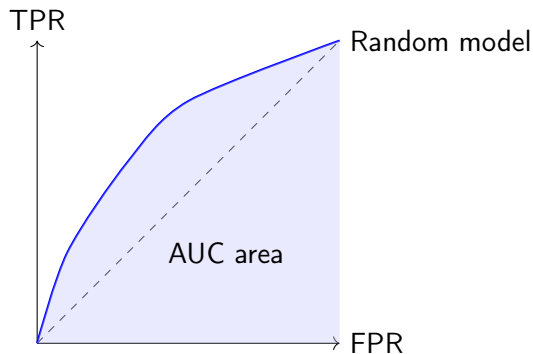
- Default threshold = 0.5
- Lower threshold $\rightarrow$ higher recall
- Higher threshold $\rightarrow$ higher precision

Choose threshold based on:

Precision / Recall trade-off + Cost of FP and FN in the domain

ROC Curve (visual)

- ROC plots TPR vs FPR for all thresholds.
- AUC measures global separability.
- AUC close to 1 = excellent model.



Advantages

- Simple, fast.
- Probabilistic output.
- Highly interpretable.
- Strong baseline model.

Limitations & best practices

- Linear decision boundary.
- Sensitive to feature scaling.

Best practices

- Always standardize continuous features.
- Tune C and penalty type.
- Inspect coefficients, F1, AUC—not just accuracy.